Image reconstruction for tomographic diffractive microscopy using equivariant imaging

Fabien Momey Casella[1], Saad Badrane[1], Flavie Benbouaziz[1], Walid Benmaarouf[1], Mélissa Corminboeuf[1], Ruben Verchère[1]

[1] *Université de Lyon, UJM-Saint-étienne, CNRS, Institut d'Optique Graduate School, Laboratoire Hubert Curien UMR 5516, F-42023, Saint-étienne, France*

fabien.momey@univ-st-etienne.fr

Tomographic Diffractive Microscopy (TDM) is a quantitative optical phase imaging technique, using coherent illumination, that enables high-resolution observation of microscopic unlabeled samples [1]. Using an image reconstruction process, it provides a 3-D map of the complex optical index of the observed sample. However, mechanical and optical constraints in the setup limit the angular coverage of the acquisition, and prevent the sample from being fully observed in terms of spatial frequencies. It results in artifacts in the TDM reconstruction, the so-called "missing cone" effect. An unsupervised deep learning technique, named Equivariant imaging (EI) [2], has been proposed to alleviate such "unseen information" issues, by enforcing invariance of the observation process to some classes of transformations.

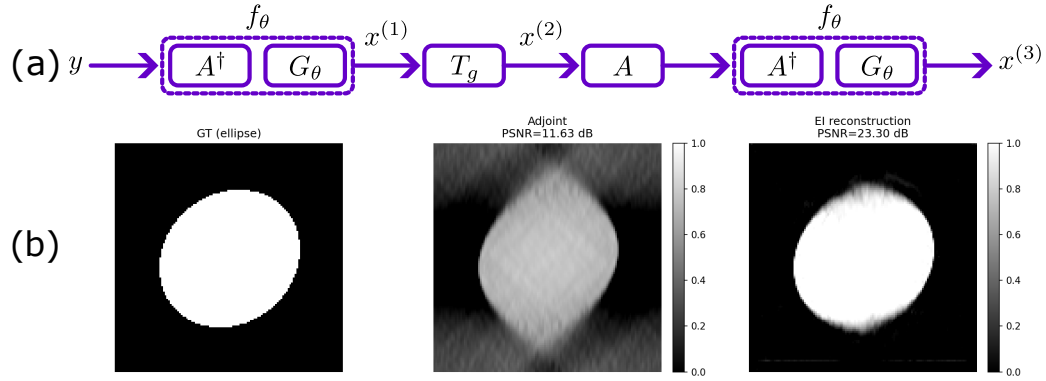


FIG. 1 : (a) Pipeline for the construction of the unsupervised EI loss. (b) Results on one ellipsoid from the test dataset compared to classical reconstruction (Born inversion).

This work presents preliminary simulation results of applying EI to TDM reconstruction. The goal is to learn an enhanced reconstructor $f_\theta = G_\theta \circ A^\dagger$ composed of a classical inversion step $A^\dagger$, followed by a trained deep neural network G_θ that have learned to eliminate the "missing cone" artifacts thanks to the EI strategy. We chose for $A^\dagger$ the reverse linear mapping from the classical linear Born model A. The unsupervised training pipeline for the loss construction is explicited in Fig. 1(a), and only requires the observed data y as input. The loss function is composed of a data-fidelity term enforcing the consistency of the "reprojected" reconstruction Ax with the data y, and an equivariance term enforcing that the reconstruction must be invariant to a defined class of transformations T_g (we chose rotations in this study). We applied this strategy to learn a TDM reconstructor on a dataset of ellipsoids of different sizes and orientations, using the PyTorch framework DeepInverse[3]. The results, shown in Fig. 1(b), are encouraging for elaborating an advanced physics-based reconstruction strategy able to mitigate the "missing cone" effect.

RÉFÉRENCES

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